Saadbin Khan

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Highlights

- 9+ years of experience in CFD simulations and multiscale heat and mass transport modeling using OpenFOAM, Python, Fortran, C++, MATLAB, ANSYS Fluent, and SolidWorks, spanning nanoscale Stokes regime to atmospheric turbulence.
- 7+ years of experience in data-driven modeling, numerical engineering analysis, multivariate analysis, non-linear regression, data processing, visualization, and interpretation.
- 5+ years of interdisciplinary research experience at the interface of computational mechanics and biological sciences.
- 4+ years of experience in SolidWorks CAD; custom meshing with *snappyHexMesh*; geometry processing and repair with Meshmixer; and data visualization and post-processing using ParaView.

Education

Ph.D. in Engineering Mechanics 2021 - 2026 (exp.)
Virginia Tech *(CGPA: 3.89)* *Blacksburg, VA*
Dissertation: The role of hydrodynamic slip and fine tracheal geometry in insect respiratory flow.
Notable courses: Finite Element Method, Fluid & Solid Mechanics, Continuum Mechanics, Intermediate Dynamics.

M.S. in Mechanical and Aerospace Engineering 2017 - 2019
Oklahoma State University *Stillwater, OK*
Thesis: Near wall structure and deviation from equilibrium in wavy channel turbulence.
Notable courses: CFD, Turbulence, Numerical Engineering Analysis, Applied Computing, Linear & Non-linear Programming.

B.Sc. in Mechanical Engineering 2011 - 2016
Bangladesh University of Engineering and Technology *Dhaka, Bangladesh*
Thesis: Mixed convection & MHD effect analysis in a lid-driven cavity with sinusoidally curved bottom wall using CNT-water nanofluid in a porous medium.
Notable courses: Thermodynamics, Heat Transfer, Mechanical Drawing, Machine Design, IC Engines, Fluidics.

Technical Strengths

Programming:	Python, Fortran, MATLAB, C++, C, Mathematica
Design & Simulation:	SolidWorks, OpenFOAM, ANSYS Fluent, COMSOL, OpenLB, Open MPI
Database & Post-processing:	Tecplot, Paraview, Microsoft Excel, Tecplot, Tracker, ImageJ
Hands-On:	Leica RM2125RTS rotary microtome

Notable Projects

◇ **The role of hydrodynamic slip and fine tracheal geometry in insect respiratory flow [Article:]** 2022 - 2026
- Funder: NSF | PI: Anne Staples | Tools: OpenFOAM, C++, Python, SolidWorks, OpenLB, Paraview
- Outcome: Demonstrated that slip and fine wall structures enhance tracheal flow beyond classical no-slip predictions; consequently developed a microscale-physics informed modified Murray's law applicable to microfluidic devices.

◇ **Waviness quantification of collagen fibers from simple harmonic generation (SHG) images [Report:]** 2022
- Funder: BIOTRANS | PI: Raffaella De Vita | Tools: Python, CT-FIRE
- Outcome: Quantified collagen fiber straightness and orientation from 20 SHG images using CT-FIRE; evaluated software limitations, demonstrating increased failure rates for highly wavy fibers and non-uniform width or opacity.

◇ **DNS of wavy channel flow to understand the near-wall turbulence structure [Articles: ,]** 2017 - 2019
- Funder: NASA Oklahoma EPSCoR | PI: Balaji Jayaraman | Tools: Fortran, Python, Open MPI, ANSYS Fluent
- Outcome: Simulated 3D turbulent flow at $Re \approx 2800$ over sinusoidal roughness using MPI parallelization with HPC, showing significantly stronger Reynolds stress distributions and turbulence structures relative to flat-wall channel flow.

◇ **Improved atmospheric plume source inversion model for remote leak detection [Presentation:]** 2018
- Funder: Baker Hughes General Electric | PI: Jamey Jacob | Tools: Python
- Outcome: Achieved up to 99.2% source localization accuracy using only 1% concentration measurement data in a structured 2D domain.

◇ **Heat transfer in nanofluids under mixed convection with MHD effects in a porous medium [Article:]** 2016
- PI: Mohammad Arif Hasan Mamun | Tools: COMSOL, Tecplot
- Outcome: Identified optimal heat transfer performance at 15% waviness to cavity width ratio and 9% CNT-nanoparticle concentration, with consistent increase in average Nusselt number across operating conditions

Relevant Experience

◇ **Research Assistant, Laboratory for Fluid Dynamics in Nature** 2021 - Present
Virginia Tech *Blacksburg, VA*

- Simulated 3D CFD models of insect respiratory transport to quantify the effects of slip and fine tracheal structures.
- Developed a microscale-physics informed modified Murray's law that applies to biological transport networks.
- 1 *Nature Reviews Physics* paper (March '25 cover article and featured in Virginia Tech newsletter), 1 *Nature Physics* paper in final phase, 1 other journal paper in review, 9 talks in conferences & symposiums.

◇ **Instructor of Record, Department of Mechanical Engineering** 2025
Virginia Tech *Blacksburg, VA*

- Courses taught: Thermodynamics (ME-2134).

◇ **Teaching Assistant** 2017 - 2019; 2021 - Present
Oklahoma State University; Virginia Tech *Stillwater, OK; Blacksburg, VA*

- Courses assisted: Thermodynamics, Fluid Mechanics–Control Volumes, Dynamics, Engineering Analysis of Physiological Systems, Engineering Analysis & Methods.

◇ **Research Assistant, Biological Transport Program** 2021 - 2022
Virginia Tech *Blacksburg, VA*

- Characterized uterosacral ligaments using straightness and orientation metrics using SHG image stacks in CT-FIRE, evaluated software performance, and identified failure modes for highly wavy and non-uniform fibers.
- Operationalized a Leica RM2125RTS rotary microtome and established protocols for repeatable tissue sectioning.

◇ **Research Assistant, Computational & Flow Science Research Laboratory** 2017 - 2019
Oklahoma State University *Stillwater, OK*

- Developed Fortran subroutines in the `Incompact3d` framework for non-flat surface flows; developed Python post-processing pipelines; performed data-driven analysis for LES-relevant turbulence modeling.
- Implemented a Python-based forward atmospheric dispersion model using 3D unsteady advection–diffusion equations to improve upon traditional Gaussian dispersion models; developed a sparse-reconstruction-based plume source inversion model in Python for remote leak detection using very limited concentration data.
- 2 journal papers, 1 symposium talk, 1 code partially utilized in BHGE's 'LUMEN' project.

◇ **Engineering Intern** 2016
Bangladesh Atomic Energy Commission *Dhaka, Bangladesh*

- Inspected structural safety, control-rod operation, and cooling water regulation in a 3 MW nuclear research reactor.

◇ **Other Services**

- **Journal Reviewer** for *Fluids*, MDPI; *Partial Differential Equations in Applied Mathematics*, Elsevier.
- **Webmaster** for the leadership team in Graduate Engineering Mechanics Society, Virginia Tech, Blacksburg, VA, USA.
- **Student Administrative Assistant** in Biological Transport program, Virginia Tech, Blacksburg, VA, USA.
- **Public Outreach Manager** in Bangladeshi Students Association, Oklahoma State University, Stillwater, OK, USA

Selected Peer-Reviewed Publications [Full list: **Google Scholar** (280+ citations)]

- [**Nature Physics *(In review)***] ***S Khan***, AE Staples, *"A tidal Murray's law framework inspired by insect respiration"*.
- [**Nature Reviews Physics 2025, Cover Article**] ***S Khan***, AE Staples, *"Mechanisms of insect respiration"*. 🔗
- [**Nanoparticle-based polymer composites 2022**] ***S Khan***, MK Hossain, *"Classification & properties of nanoparticles"*. 🔗
- [**AIP Advances 2020**] B Jayaraman, ***S Khan***, *"Direct numerical simulation of turbulence over two-dimensional waves"*. 🔗
- [**Fluids 2019**] ***S Khan***, B Jayaraman, *"Statistical structure and deviations from equilibrium in wavy channel turbulence"*. 🔗
- [**AIP Conference Proceedings 2016**] MI Khan, KM Rabbi, ***S Khan***, M Mamun, *"Mixed convection analysis in lid-driven cavity with sinusoidally curved bottom wall using CNT-water nanofluid"*. 🔗

Awards

- **Alice & Dan Pletta Scholarship**, Engineering Mechanics Graduate Program, Virginia Tech 2026
- **BIOTRANS Fellowship**, Interdisciplinary Graduate Education Programs, Virginia Tech 2021 - 2022
- **Champion**, 6th and 8th Bangladesh Mathematical Olympiad 2008 & 2010

(References available upon request)